Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
3	*(a)*	(i)	Sirius A: $\lambda_{max} = 290 \times 10^{-9}$ [m] **and** Canopus: $\lambda_{max} = 400 \times 10^{-9}$ [m] [± 10 nm] for both (1) Attempt at **applying** $\lambda_{max}T = 0.0029$ to both stars, even if powers of 10 incorrect (1) No need to change unit of λ to m Correct application, either by confirming Wien constant **or** star temperatures **or** λ_{max} (1) Accept correct calculations of λ_{max} even if no reference is made to the graph Application of Wien's law to one star only award 1 mark only			3	3	2	
		(ii)	Sirius A (1) Greater spectral intensity (at 'blue' end or at shorter wavelengths or towards 400 nm) (1) Don't accept peak wavelength of Sirius A is closest to the blue end of the spectrum than Canopus. Don't accept reference to temperature by itself e.g. Sirius has a higher temperature so therefore must be bluer. Accept ratio of $\frac{B}{R}$ for both stars, e.g. $\frac{6.1}{1.7}$ against $\frac{1.6}{0.8}$		1	1	2		

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)	Correct use of $P = A\sigma T^4$ for either Sirius A **or** Canopus **or** both i.e. Canopus: $P = 4 \times \pi \times (4.97 \times 10^{10})^2 \times 5.67 \times 10^{-8} \times (7\,250)^4$ Sirius A: $P = 4 \times \pi \times (1.19 \times 10^9)^2 \times 5.67 \times 10^{-8} \times (10\,000)^4$ (1) Canopus: $P = 4.9 \times 10^{30}$ [W] (1) Sirius A: $P = 1.0 \times 10^{28}$ [W] (1) Accept powers of 10 error in answers omission of 4 deduct 1 mark $\frac{4.9\times10^{30}}{1.0\times10^{28}}$ (≈ 500) shown (1) **Alternative:** Attempt at $P = A\sigma T^4$ used in ratio (1) e.g. $\frac{L_C}{L_S} = \frac{49.7 \text{ x } 7\,250^4 (1)}{1.192 \text{ x } 10\,000 (1)}$ ≈ 500 (1)	1	1 1 1		4	4	
		(ii)	$I = \frac{1.0\times10^{28}}{4\pi (8.15\times10^{16})^2}$ substitution (1) **ecf** $I = 1.19 \times 10^{-7}$ [W m^{-2}] (1) Accept 1.21×10^{-7} [W m^{-2}]	1	1		2	2	
		(iii)	Canopus is further away from earth or Sirius A is closer to earth (1) Intensity reaching earth α $\frac{1}{R^2}$ **or** P from star spread out over greater surface area (1) Accept intensity equation Don't accept intensity α $\frac{1}{\text{distance}}$ **or** because of the inverse square law		2		2		
			Question 3 total	**2**	**7**	**4**	**13**	**8**	**0**